

UCT41042 – Practical for Introduction to Smart System
Department of Information & Communication Technology
Faculty of Technology
South Eastern University of Sri Lanka
Lab Sheet No: 05

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Title: Programming ESP32 with Sensors in Wokwi

Aims: To program the ESP32 to interface with various sensors, understand their libraries and communication protocols, and display real-time sensor data on the Serial Monitor.

Tasks:

1. HC-SR04 (Ultrasonic Distance Sensor)

Write a program to measure distance in centimeters and display it on the Serial Monitor every second.

2. DHT22 (Digital Humidity & Temperature Sensor)

Write a program to read and display both temperature (°C) and humidity (%) every 2 seconds.

3. DS18B20 (One-Wire Digital Temperature Sensor)

Write a program to read and display the temperature in Celsius.

4. BMP180 (Barometric Pressure & Temperature Sensor)

Write a program to display Temperature (°C), Pressure (hPa), and Altitude (meters).

5. MPU6050 (3-Axis Accelerometer + Gyroscope + Temperature)

Write a program to display acceleration (X, Y, Z) and gyroscope (X, Y, Z) values.

6. PIR Motion Sensor (Passive Infrared)

Write a program to detect motion and print “Motion Detected” or “No Motion” continuously.

7. Photoresistor (LDR) Light Sensor

Write a program to read and display the analog light intensity value.

8. Analog Temperature Sensor (NTC Thermistor)

Write a program to calculate and display temperature in °C using the thermistor.

9. MQ2 Gas Sensor

Write a program to read and display the gas/smoke level (analog value) every 500ms.

10. HX711 Load Cell Amplifier

Write a program to display weight in grams. Include taring (zeroing) at startup.

11. MFRC522 RFID/NFC Reader

Write a program to read an RFID card and print its UID on the Serial Monitor.